

STEP-2

For JEE (Main) & State Engineering Entrance Exams

Assignment

* Questions to be discussed in Class.

Objective Type Questions (One option is correct)

1. Identify the correct statement.

- (1) Distance travelled represents the magnitude of displacement
- (2) Average speed represents the magnitude of average velocity
- (3) Average acceleration = $\frac{\text{Average velocity}}{\text{Time}}$
- (4) Instantaneous speed = magnitude of instantaneous velocity

2. A body moves from $x = -1$ to $x = 4$ and then to $x = -4$. Which of the following is correct?

- (1) Its displacement is zero
- (2) The distance covered by it is 5 units
- (3) Its displacement is -3
- (4) Distance covered by it is 9 units

3. The acceleration of a particle is given by the $a = x$ where x is a constant. If the particle starts at origin from rest, its distance from origin after time t is given by

- (1) $\frac{xt^2}{2}$
- (2) $\frac{x^2t^2}{2}$
- (3) $\frac{\sqrt{x}t^2}{2}$
- (4) $\frac{x^3t^3}{6}$

4. The relation between the time t and position x for a particle moving on x -axis is given by $t = px^2 + qx$, where p and q are constants. The relation between velocity v and acceleration a is as

- (1) $a \propto v^3$
- (2) $a \propto v^2$
- (3) $a \propto v^4$
- (4) $a \propto v$

5. Two balls X and Y are thrown from top of tower one vertically upward and other vertically downward with same speed. If times taken by them to reach the ground are 6 s and 2 s respectively, then the height of the tower and initial speed of each ball are ($g = 10 \text{ m/s}^2$)

- (1) 60 m, 15 m/s
- (2) 80 m, 20 m/s
- (3) 60 m, 20 m/s
- (4) 45 m, 10 m/s

Paragraph for Q. Nos. 6 to 11

A ball A is thrown up vertically with speed u . At the same instant another ball B is released from rest at a height h vertically above the point of projection of A . Taking the moment of projection as $t = 0$ and acceleration due to gravity is ' g ', answer the following questions.

6. The separation between the bodies at any time instant t is

- (1) h
- (2) $h - ut$
- (3) $h - ut + \frac{gt^2}{2}$
- (4) $h - ut + gt^2$

7. The time instant at which they collide is

- (1) $\frac{h}{u}$
- (2) $\frac{2h}{u}$
- (3) $\sqrt{\frac{2h}{g}}$
- (4) $\frac{2h}{g}$

8. At the moment of collision, the position of A is at a distance

- (1) h above ground
- (2) $h - \frac{gh^2}{2u^2}$ above ground
- (3) $\frac{gh^2}{2u^2}$ below the point of release of B
- (4) Both (2) & (3) are correct

9. At the moment of collision, the position of B is at a distance

- (1) h above ground
- (2) $h - \frac{gh^2}{2u^2}$ above ground
- (3) $\frac{gh^2}{2u^2}$ below the point of release of B
- (4) Both (2) & (3) are correct

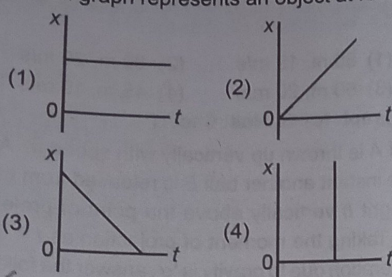
10. At any time instant t , the magnitude of relative acceleration of A with respect to B is

(1) g (2) $2g$
(3) $-2g$ (4) Zero

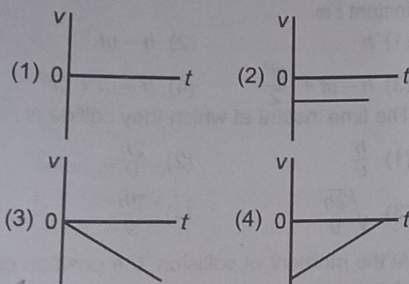
11. At any time instant t , the speed of A relative to B is

(1) u (2) $u - 2gt$
(3) $\sqrt{u^2 - 2gh}$ (4) $u - gt$

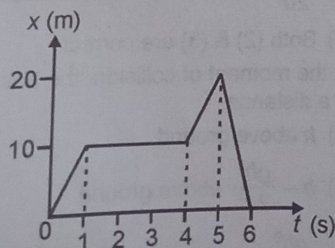
12. Which graph represents an object at rest?



13. Which graph represents constant positive acceleration?

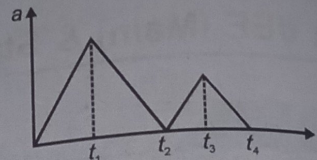


14. Figure shows the graph of x -coordinate of a particle moving along x -axis as a function of time. Average velocity during $t = 0$ to 4 s and instantaneous velocity at $t = 4.113$ s respectively will be



- (1) 5 m/s, 10 m/s (2) 2.5 m/s, 10 m/s
(3) Zero, zero (4) 10 m/s, 2.5 m/s

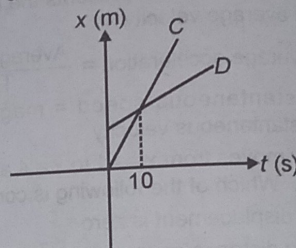
15. A particle starts moving from rest on a straight line. Its acceleration a versus time t is shown in the figure. The speed of the particle is maximum at the instant



- (1) t_1 (2) t_2
(3) t_3 (4) t_4

Paragraph for Q. Nos. 16 to 19

The figure shows position-time graph of two riders C and D. Based on the information represented by the graph, answer the following questions.



16. At $t = 0$ s

(1) Rider C is ahead of rider D
(2) Rider D is ahead of rider C
(3) Rider C and D are at the same position
(4) Not enough information is given

17. At $t = 0$ s

(1) C is moving, and D is at rest
(2) D is moving, and C is at rest
(3) C and D are both moving
(4) C and D are both at rest

18. At $t = 0$ s

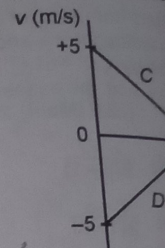
(1) C has a greater velocity than D
(2) D has a greater velocity than C
(3) C and D have the same velocity
(4) C is accelerating

19. At $t = 10$ s

(1) C and D are at the same position
(2) C and D have the same velocity
(3) The velocity of D is greater than the velocity of C
(4) C is in front of D

Paragraph for Q. Nos. 20 to 22

The figure shows velocity-time graph of two cars C and D. Based on the information represented by the graph, answer the following questions.



20. During the first 8 s

(1) C has decelerating motion
(2) C and D both have the same velocity
(3) C and D have the same acceleration
(4) C has the same velocity as D

21. Based on all the information

(1) They meet at $t = 10$ s
(2) They will meet at $t = 10$ s
(3) They will meet at $t = 10$ s
(4) Not enough information is given

22. The acceleration of car C is

(1) 12 s
(2) 10 s
(3) 8 s
(4) 6 s

Paragraph for Q. Nos. 23 to 25

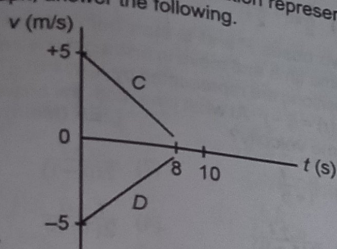
Two cars start from rest with the same acceleration of 10 m/s^2 and move in the same direction. The first car starts at $t = 0$ s and the second car starts at $t = 10$ s. Based on the information represented by the graph, answer the following questions.

23. As seen from the first car, the second car

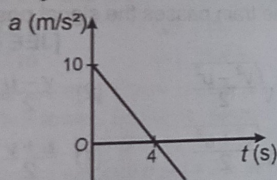
(1) Car is at rest
(2) First car is moving
(3) Second car is moving
(4) Both cars are moving

Paragraph for Q. Nos. 20 and 21

The figure shows velocity-time graph of two riders C and D. Based on the information represented by the graph, answer the following.



20. During the first 8 s
- (1) C has decreasing speed and D has increasing speed
 - (2) C and D both have decreasing speed
 - (3) C and D have the same velocity
 - (4) C has the same average velocity as D
21. Based on all the graphical information
- (1) They meet at the same position at $t = 8$ s
 - (2) They will meet at the same position, at $t = 10$ s
 - (3) They will never meet at the same position
 - (4) Not enough information is given to decide if they meet
22. The acceleration-time graph of a particle is as shown. At what time the particle acquires its initial velocity?



- (1) 12 s
- (2) 5 s
- (3) 8 s
- (4) 16 s

Paragraph for Q. Nos. 23 to 30

Two cars start from same point on a straight road due North. While car A starts with an initial velocity of 10 m/s and an acceleration of 1 m/s^2 , car B starts from rest with an acceleration of 2 m/s^2 . Answer the following questions.

23. As seen by a passenger in car A,
- (1) Car B moves backward
 - (2) First, the car B moves backward and then forward

- (3) Car B moves forward
 - (4) First, the car B moves forward and then backward
24. As seen by a passenger in car B,
- (1) Car A moves backward
 - (2) First, the car A moves backward and then forward
 - (3) Car A moves forward
 - (4) First, the car A moves forward and then backward

25. The separation between the cars
- (1) Continuously increases
 - (2) Continuously decreases
 - (3) First increases, then decreases and finally increases again
 - (4) First decreases, then increases and finally decreases again

26. Taking the moment of start as $t = 0$, the separation between the cars at the moment $t = 5$ s is

- (1) 37.5 m with A ahead of B
- (2) 37.5 m with B ahead of A
- (3) 50 m with B ahead of A
- (4) 50 m with A ahead of B

27. The position of car A at this moment ($t = 5$ s) is

- (1) 62.5 m due North of starting point
- (2) 37.5 m due North of starting point
- (3) 62.5 m due South of starting point
- (4) 37.5 m due South of starting point

28. The position of car B at this moment ($t = 5$ s) is

- (1) 25 m due North of starting point
- (2) 37.5 m due North of starting point
- (3) 25 m due South of starting point
- (4) 37.5 m due South of starting point

29. The moment of time at which the separation between the cars is same as that at $t = 5$ s, is given by $t =$

- (1) 15 s
- (2) 20 s
- (3) 10 s
- (4) 7 s

30. The separation between the cars is zero at the moment $t = 0$. The moment of time at which the separation again becomes zero is $t =$

- (1) 10 s
- (2) 20 s
- (3) 25 s
- (4) 35 s

Numerical Value Based Questions

31. A particle travels half the distance of a straight journey with a speed 5 m/s. The remaining part of the distance is covered with speed 6 m/s for half the remaining time, and with speed 4 m/s for the other half of the remaining time. The average speed of the particle is n m/s. The value of n is _____.
32. A particle moves along a straight line such that its displacement at any instant ' t ' is given by $x = t^3 - 6t^2 + 3t + 4$ meters. The velocity when the acceleration is zero is n m/s. The value of n is _____.
33. A body starts from rest with an acceleration 2 m/s^2 till it attains the maximum velocity then retards to rest with 3 m/s^2 . If total time taken is 10 second, then maximum speed attained is n m/s. The value of n is _____.
34. A body is thrown vertically upward with velocity u . The distance travelled by it in the 7th and 8th seconds are equal. The displacement in 8th seconds is equal to n m. The value of n is _____. (take $g = 10 \text{ m/s}^2$)
35. A motor boat going downstream overcomes a float at a point A. 60 minutes later it turns and after some time passes the float at a distance of 12 km from the point A. The velocity of the stream is $n \text{ km/h}$. The value of n is _____. (assuming constant velocity for the boat in still water)

PREVIOUS YEARS QUESTIONS

Objective Type Questions

[Equation of Motion]

36. The position of a particle as a function of time t , is given by
 $x(t) = at + bt^2 - ct^3$
 where a , b and c are constants. When the particle attains zero acceleration, then its velocity will be [JEE (Main)-2019]
- (1) $a + \frac{b^2}{4c}$ (2) $a + \frac{b^2}{3c}$
 (3) $a + \frac{b^2}{2c}$ (4) $a + \frac{b^2}{c}$
37. The instantaneous velocity of a particle moving in a straight line is given as $v = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1 s and 2 s is [JEE (Main)-2021]

- (1) $\frac{3}{2}\alpha + \frac{7}{3}\beta$ (2) $\frac{\alpha}{2} + \frac{\beta}{3}$
 (3) $\frac{3}{2}\alpha + \frac{7}{2}\beta$ (4) $3\alpha + 7\beta$

38. Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity? [JEE (Main)-2022]

- (1) $\frac{\alpha - f}{1 + \beta}$ (2) $\frac{\alpha + f}{2(\beta - 1)}$
 (3) $\frac{\alpha + f}{2(1 + \beta)}$ (4) $\frac{f - \alpha}{2(1 + \beta)}$

39. A helicopter rises from rest on the ground vertically upwards with a constant acceleration g . A food packet is dropped from the helicopter when it is at a height h . The time taken by the packet to reach the ground is close to $[g$ is the acceleration due to gravity]

[JEE (Main)-2020]

- (1) $t = 3.4\sqrt{\left(\frac{h}{g}\right)}$ (2) $t = 1.8\sqrt{\left(\frac{h}{g}\right)}$
 (3) $t = \sqrt{\frac{2h}{3g}}$ (4) $t = \frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$

40. An engine of a train, moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v . The velocity with which middle point of the train passes the signal post is:

[JEE (Main)-2021]

- (1) $\sqrt{\frac{v^2 - u^2}{2}}$ (2) $\frac{v - u}{2}$
 (3) $\sqrt{\frac{v^2 + u^2}{2}}$ (4) $\frac{u + v}{2}$

41. A stone is dropped from the top of a building. When it crosses a point 5 m below the top, another stone starts to fall from a point 25 m below the top. Both stones reach the bottom of building simultaneously. The height of the building is [JEE (Main)-2021]

- (1) 35 m (2) 45 m
 (3) 25 m (4) 50 m

42. A car accelerates from rest at a constant rate α for some time after which it decelerates at a constant rate β to come to rest. If the total time elapsed is t seconds, the total distance travelled is: [JEE (Main)-2021]

- (1) $\frac{2\alpha\beta}{(\alpha+\beta)}t^2$ (2) $\frac{\alpha\beta}{2(\alpha+\beta)}t^2$
 (3) $\frac{\alpha\beta}{4(\alpha+\beta)}t^2$ (4) $\frac{4\alpha\beta}{(\alpha+\beta)}t^2$

43. A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1 . If he remains stationary on a moving escalator then the escalator takes him up in time t_2 . The time taken by him to walk up on the moving escalator will be:

[JEE (Main)-2021]

- (1) $t_2 - t_1$ (2) $\frac{t_1 t_2}{t_2 - t_1}$
 (3) $\frac{t_1 + t_2}{2}$ (4) $\frac{t_1 t_2}{t_2 + t_1}$

44. A balloon was moving upwards with a uniform velocity of 10 m/s. An object of finite mass is dropped from the balloon when it was at a height of 75 m from the ground level. The height of the balloon from the ground when object strikes the ground was around (takes the value of g as 10 m/s²)

[JEE (Main)-2021]

- (1) 250 m (2) 125 m
 (3) 300 m (4) 200 m

45. When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v . It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is:

[JEE (Main)-2022]

- (1) 19.6 m (2) 29.4 m
 (3) 39.2 m (4) 73.5 m

46. Two balls A and B are placed at the top of 180 m tall tower. Ball A is released from the top at $t = 0$ s. Ball B is thrown vertically down with an initial velocity u at $t = 2$ s. After a certain time, both balls meet 100 m above the ground. Find the value of u in ms⁻¹. [use $g = 10$ ms⁻²]

[JEE (Main)-2022]

- (1) 10 (2) 15
 (3) 20 (4) 30

47. A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10 m in t s, the distance travelled by the toy in the next t s will be:

[JEE (Main)-2022]

- (1) 10 m (2) 20 m
 (3) 30 m (4) 40 m

48. A NCC parade is going at a uniform speed of 9 km/h under a mango tree on which a monkey is sitting at a height of 19.6 m. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is

(Given $g = 9.8$ m/s²)

[JEE (Main)-2022]

- (1) 5 m (2) 10 m
 (3) 19.8 m (4) 24.5 m

49. A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h . Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively

[JEE (Main)-2022]

- (1) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (2) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$
 (3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{1}{3}$

50. A car travels a distance of 'x' with speed v_1 and then same distance 'x' with speed v_2 in the same direction. The average speed of the car is

[JEE (Main)-2023]

- (1) $\frac{v_1 + v_2}{2}$ (2) $\frac{v_1 v_2}{2(v_1 + v_2)}$
 (3) $\frac{2v_1 v_2}{v_1 + v_2}$ (4) $\frac{2x}{v_1 + v_2}$

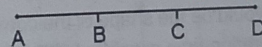
51. The distance travelled by a particle is related to time t as $x = 4t^2$. The velocity of the particle at $t = 5$ s is:

[JEE (Main)-2023]

- (1) 8 ms⁻¹ (2) 20 ms⁻¹
 (3) 40 ms⁻¹ (4) 25 ms⁻¹

52. An object moves with speed v_1 , v_2 and v_3 along a line segment AB, BC and CD respectively as shown in figure. Where $AB = BC$ and $AD = 3AB$, then average speed of the object will be:

[JEE (Main)-2023]



- (1) $\frac{v_1 v_2 v_3}{3(v_1 v_2 + v_2 v_3 + v_3 v_1)}$
 (2) $\frac{(v_1 + v_2 + v_3)}{3}$
 (3) $\frac{3v_1 v_2 v_3}{(v_1 v_2 + v_2 v_3 + v_3 v_1)}$
 (4) $\frac{(v_1 + v_2 + v_3)}{3v_1 v_2 v_3}$

66 Motion in a Straight Line

53. Position of an ant (S in metres) moving in $Y-Z$ plane is given by $S = 2t^2\hat{j} + 5t\hat{k}$ (where t is in second). The magnitude and direction of velocity of the ant at $t = 1$ s will be

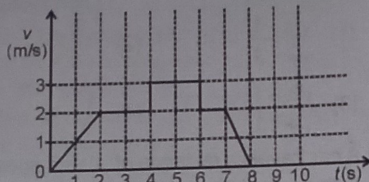
[JEE (Main)-2024]

- (1) 4 m/s in y -direction (2) 9 m/s in z -direction
(3) 16 m/s in y -direction (4) 4 m/s in x -direction

[Graphical Analysis]

54. A particle starts from the origin at time $t = 0$ and moves along the positive x -axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle time $t = 5$ s?

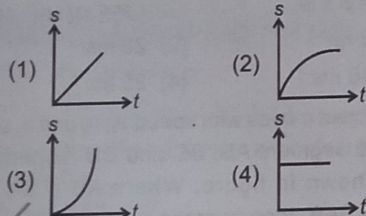
[JEE (Main)-2019]



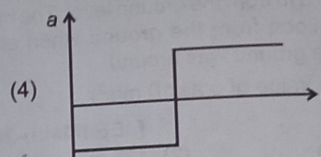
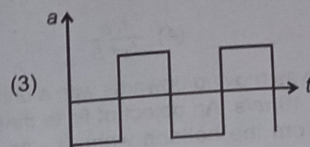
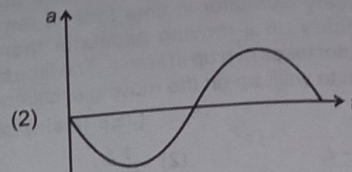
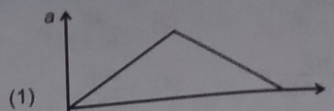
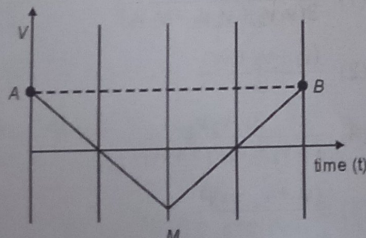
- (1) 9 m (2) 6 m
(3) 10 m (4) 3 m

55. A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) - time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale)

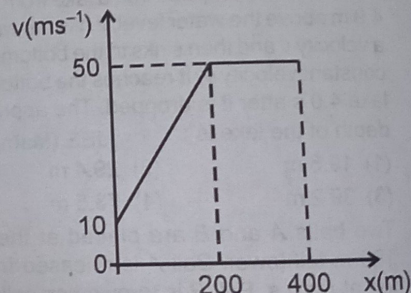
[JEE (Main)-2020]



56. If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph? [JEE (Main)-2021]

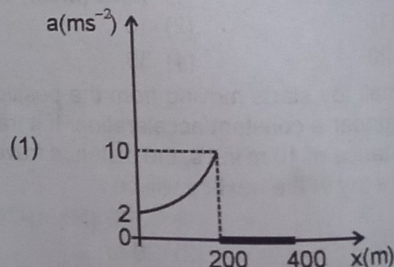


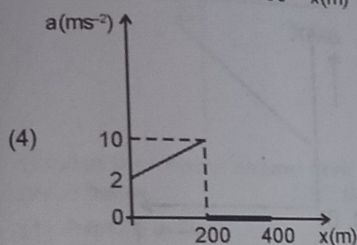
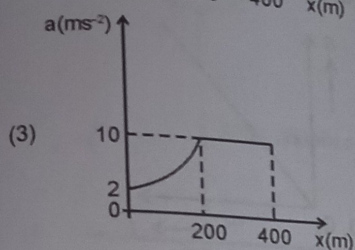
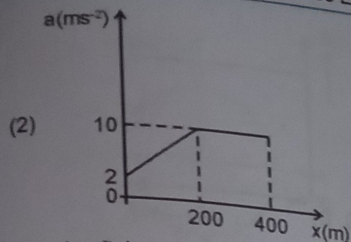
57. The velocity-displacement graph describing the motion of a bicycle is shown in the figure.



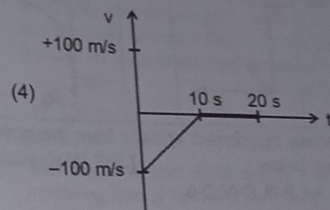
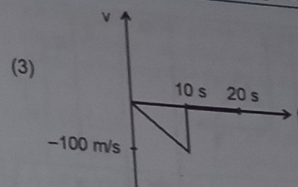
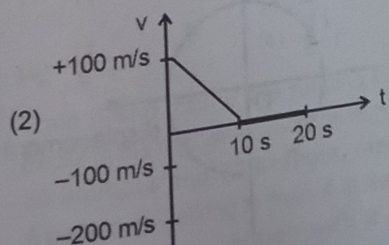
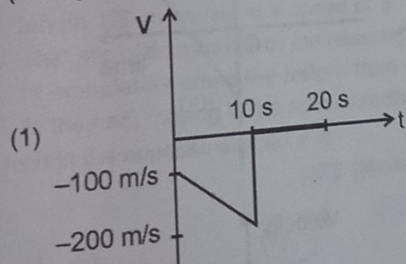
The acceleration-displacement graph of the bicycle's motion is best described by:

[JEE (Main)-2021]

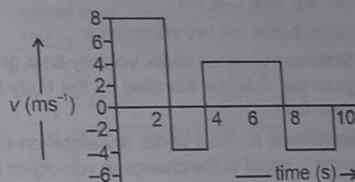




58. A bullet is shot vertically downwards with an initial velocity of 100 m/s from a certain height. Within 10 s, the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity-time curve for total time $t = -20$ s will be (Take $g = 10 \text{ m/s}^2$) [JEE (Main)-2022]



59. The velocity-time graph of a body moving in a straight line is shown in figure.



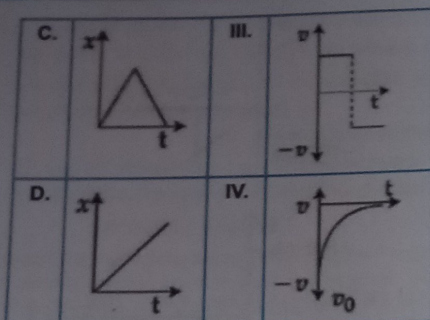
The ratio of displacement to distance travelled by the body in time 0 to 10 s is :

[JEE (Main)-2023]

- (1) 1 : 4 (2) 1 : 2
(3) 1 : 3 (4) 1 : 1

60. Match Column-I with Column-II :

	Column-I (x-t graphs)	Column-II (v-t graphs)
A.		I.
B.		II.



Choose the correct answer from the options given below : [JEE (Main)-2023]

- (1) A-I, B-III, C-IV, D-II
- (2) A-II, B-III, C-IV, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-IV, C-III, D-I

61. Given below are two statements:

Statement I: Area under velocity-time graph gives the distance travelled by the body in a given time.

Statement II: Area under acceleration-time graph is equal to the change in velocity-in the given time.

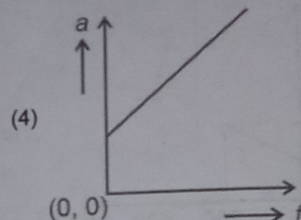
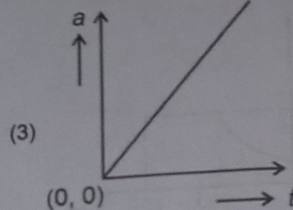
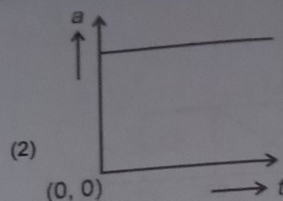
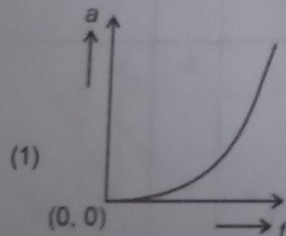
In the light of given statements, choose the correct answer from the options given below.

[JEE (Main)-2023]

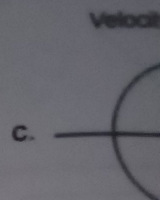
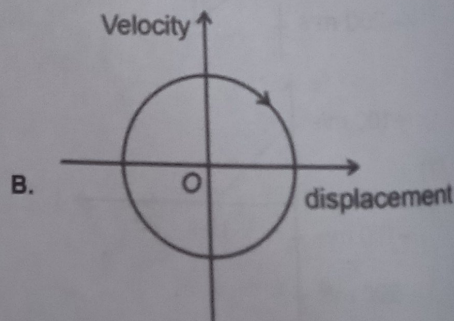
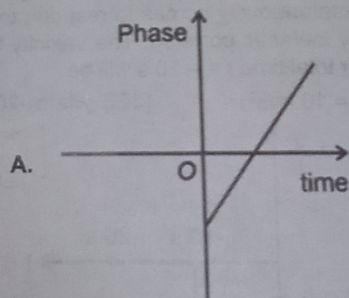
- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is correct but Statement II is false
- (4) Statement I is incorrect but Statement II is true

62. A wooden block, initially at rest on the ground, is pushed by a force which increases linearly with time t . Which of the following curve best describes acceleration of the block with time?

[JEE (Main)-2024]



63. Which of the following curves possibly represent one-dimensional motion of a particle?



D.

Choose the given below

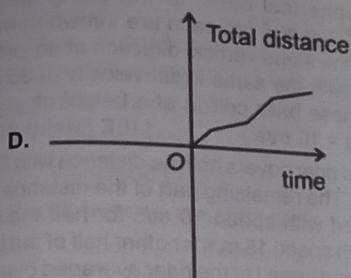
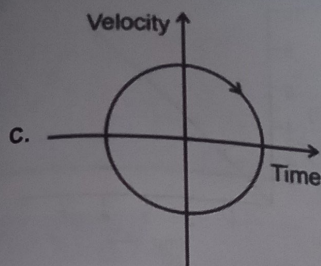
- (1) A and B
- (2) A, C and D
- (3) A, B and C
- (4) A, B and D

64. A passenger train is moving with a speed of 120 km/h. The ratio of the length of the train to the length of the platform is (i) they are equal (ii) in the ratio of 1:1

(1) $\frac{25}{11}$

(3) $\frac{11}{5}$

65. Train A is moving with a speed of 36 km/h. A person is walking in the opposite direction to the train.



Choose the correct answer from the options given below :

- (1) A and B only
(2) A, C and D only
(3) A, B and C only
(4) A, B and D only

[Relative Motion]

64. A passenger train of length 60 m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/hr. The ratio of times taken by the passenger train to completely cross the freight train when :
(i) they are moving in the same direction, and
(ii) in the opposite directions is

[JEE (Main)-2019]

- (1) $\frac{25}{11}$ (2) $\frac{5}{2}$
(3) $\frac{11}{5}$ (4) $\frac{3}{2}$

65. Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 km/hour and 72 km/hour, respectively. A person is walking in train A in the direction opposite to its motion with a speed of

1.8 km/hour. Speed (in ms^{-1}) of this person as observed from train B will be close to (take the distance between the tracks as negligible)

[JEE (Main)-2020]

- (1) 30.5 ms^{-1} (2) 29.5 ms^{-1}
(3) 31.5 ms^{-1} (4) 28.5 ms^{-1}

66. A passenger sitting in a train A moving at 90 km/h observes another train B moving in the opposite direction for 8 s. If the velocity of the train B is 54 km/h, then length of train B is:

[JEE (Main)-2023]

- (1) 120 m (2) 320 m
(3) 80 m (4) 200 m

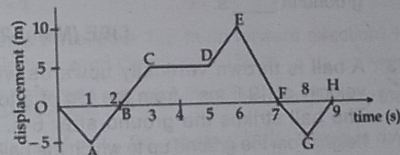
67. Two trains 'A' and 'B' of length 'l' and '4l' are travelling into a tunnel of length 'L' in parallel tracks from opposite directions with velocities 108 km/h and 72 km/h, respectively. If train 'A' take 35 s less time than train 'B' to cross the tunnel then, length 'L' of tunnel is:

(Given $L = 60 \text{ l}$)

[JEE (Main)-2023]

- (1) 1200 m (2) 900 m
(3) 1800 ms (4) 2700 m

68. The displacement x versus time graph is shown below.



- (A) The average velocity during 0 to 3 s is 10 m/s
(B) The average velocity during 3 to 5 s is 0 m/s
(C) The instantaneous velocity at $t = 2 \text{ s}$ is 5 m/s
(D) The average velocity during 5 to 7 s and instantaneous velocity at $t = 6.5 \text{ s}$ are equal
(E) The average velocity from $t = 0$ to $t = 9 \text{ s}$ is zero

Choose the correct answer from the options given below

[JEE (Main)-2025]

- (1) (B), (D), (E) only
(2) (B), (C), (D) only
(3) (A), (D), (E) only
(4) (B), (C), (E) only

70 Motion in a Straight Line

Numerical Value Based Questions

69. The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x^{-n} , where n is an integer, the value of n is _____.

[JEE (Main)-2020]

70. If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x}$ m/s, then the acceleration of the body is _____ m/s².

[JEE (Main)-2021]

71. A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2}$ s before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms⁻²) near the surface on that planet is _____.

[JEE (Main)-2020]

72. From the top of a tower, a ball is thrown vertically upward which reaches the ground in 6 s. A second ball thrown vertically downward from the same position with the same speed reaches the ground in 1.5 s. A third ball released, from the rest from the same location, will reach the ground in _____ s.

[JEE (Main)-2022]

- 73*. A ball is thrown vertically upwards with a velocity of 19.6 ms^{-1} from the top of a tower. The ball strikes the ground after 6 s. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)$ m. The value of k is _____.

(use $g = 9.8 \text{ m/s}^2$)

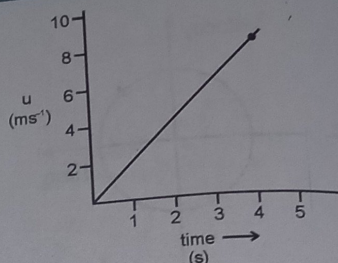
[JEE (Main)-2022]

74. A ball is projected vertically upward with an initial velocity of 50 ms^{-1} at $t = 0$ s. At $t = 2$ s, another ball is projected vertically upward with same velocity. At $t =$ _____ s, second ball will meet the first ball. ($g = 10 \text{ ms}^{-2}$)

[JEE (Main)-2022]

75. The speed versus time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval $t = 0$ to $t = 5$ s will be _____.

[JEE (Main)-2020]



76. Two spherical balls having equal masses with radius of 5 cm each are thrown upwards along the same vertical direction at an interval of 3 s with the same initial velocity of 35 m/s, then these balls collide at a height of _____ m. (take $g = 10 \text{ m/s}^2$)

[JEE (Main)-2021]

77. A horse rider covers half the distance with 5 m/s speed. The remaining part of the distance was travelled with speed 10 m/s for half the time and with speed 15 m/s for other half of the time. The mean speed of the rider averaged over the whole time of motion is $\frac{x}{7}$ m/s. The value of x is _____.

[JEE (Main)-2023]

78. A body moves on a frictionless plane starting from rest. If S_n is distance moved between $t = n - 1$ and $t = n$ and S_{n-1} is distance moved between $t = n - 2$ and $t = n - 1$, then the ratio $\frac{S_{n-1}}{S_n}$ is $\left(1 - \frac{2}{x}\right)$ for $n = 10$. The value of x is _____.

[JEE (Main)-2024]

79. Two cars P and Q are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time $t = 0$, for the first time. The maximum possible number of crossing(s) (including the crossing at $t = 0$) is _____.

[JEE (Main)-2025]

80. A person travelling on a straight line moves with a uniform velocity v_1 for a distance x and with a uniform velocity v_2 for the next $\frac{3}{2}x$ distance.

The average velocity in this motion is $\frac{50}{7}$ m/s.If v_1 is 5 m/s then $v_2 =$ _____ m/s.

[JEE (Main)-2025]

Example 1 : A particle starts from rest with a constant acceleration. Its position in meters by the time t is given by the equation $x = at^3$. The magnitude of its acceleration is _____.

- Initial velocity
- Velocity
- Uniform
- Position

Solution : $v = \frac{dx}{dt}$

- For initial velocity

$$\therefore V_{\text{initial}} = 4$$

- When, $t = 3$

$$\text{velocity} = 4$$

- Acceleration

$$a = \frac{dv}{dt}$$

$$\therefore a = 10$$

- At $t = 5$

$$x = 3$$

$$= 3$$

$$= 1$$

Example 2 :

that its position x is given by the relation $x = 5t^3$. The distance travelled in the first 5 seconds is _____.

Solution : $x = 5t^3$

$$= [50 \times 0 - 50 \times (-5)]$$

Distance travelled

straight line

particle does

a turning point

becomes zero

Corporate

